

Section 1: Array Assignments (1-20)

1.

Create an array of 10 employee names and display them using `foreach`.

2.

Find the largest number from an array.

3.

Find the smallest number from an array.

4.

Calculate the sum of all array elements.

5.

Calculate the average of array values.

6.

Count how many even numbers are present in an array.

7.

Count how many odd numbers are present in an array.

8.

Remove duplicate values from an array.

9.

Sort an array in ascending order.

10.

Sort an array in descending order.

11.

Merge two arrays and remove duplicates.

12.

Find whether a particular value exists in an array.

13.

Reverse an array without using `array_reverse()`.

14.

Create an associative array of students and marks and find the topper.

15.

Convert a comma-separated string into an array.

16.

Convert an array into a comma-separated string.

17.

Find the second highest number in an array.

18.

Find all duplicate elements in an array.

19.

Create a multidimensional array of products and display product names with prices.

20.

Group products by category using arrays.

Section 2: Loop Assignments (21-35)

21.

Print numbers from 1 to 100 using a `for` loop.

22.

Print even numbers from 1 to 100.

23.

Print odd numbers from 1 to 100.

24.

Calculate the factorial of a number.

25.

Generate the multiplication table of a given number.

26.

Print the Fibonacci series.

27.

Print the following pattern:

```
*  
* *  
* * *  
* * * *  
* * * * *
```

28.

Print the following pattern:

```
* * * * *  
* * * *  
* * *  
* *  
*
```

29.

Print a pyramid pattern.

30.

Count the number of digits in a given number.

31.

Reverse a number.

32.

Check whether a number is a palindrome.

33.

Check whether a number is an Armstrong number.

34.

Find all prime numbers between 1 and 100.

35.

Create a menu-driven calculator using loops.

Section 3: Function Assignments (36-50)

36.

Create a function to add two numbers.

37.

Create a function to find the factorial of a number.

38.

Create a function to check whether a number is prime.

39.

Create a function to reverse a string.

40.

Create a function to count vowels in a string.

41.

Create a function to find the maximum value from an array.

42.

Create a function that accepts an array and returns the sum of elements.

43.

Create a function to generate a random password.

44.

Create a function to calculate age from date of birth.

45.

Create a recursive function to print numbers from 1 to N.